Cohort canonical evaluation — 60 agents

idea-01-lateral-attribution · 534 held-out segments · V0 yaw 0.01627 rad/s, CTE 254.26 m · generated 2026-06-01 17:38

Executive summary

AGENTS (OK / TOTAL)

59 / 60

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

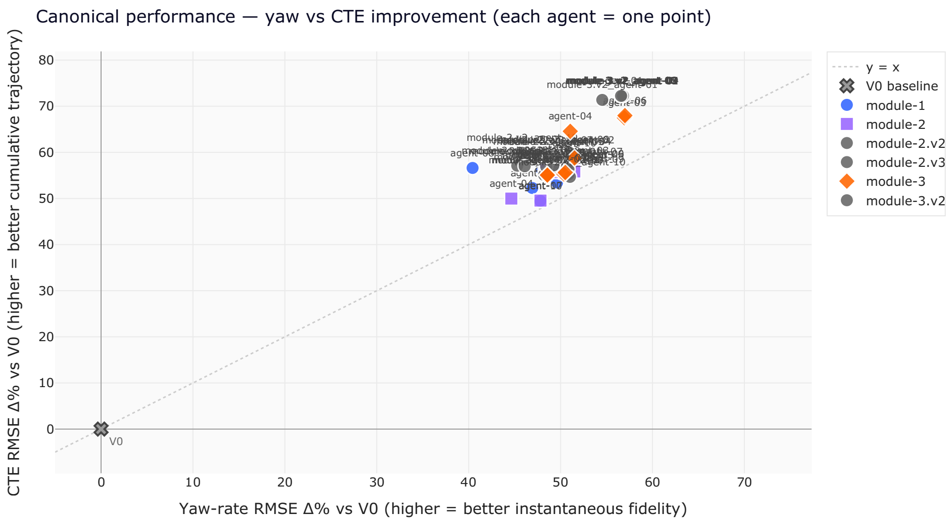
24.17s

Best yaw: `m3-agent-06` **+57.0%** · Best CTE: `module-3.v2_agent-04` **+72.3%**

Winning both KPIs ≥ +30% (59 agents): `m3-agent-06`, `m3-agent-03`, `module-3.v2_agent-04`, `module-3.v2_agent-09`, `module-3.v2_agent-02`, `module-3.v2_agent-03`, `m3-agent-01`, `module-3.v2_agent-05`, `module-3.v2_agent-06`, `module-3.v2_agent-07`, `module-3.v2_agent-08`, `module-3.v2_agent-10`, `module-3.v2_agent-01`, `m3-agent-10`, `m2-agent-05`, `m3-agent-04`, `module-2.v3_agent-10`, `m1-agent-01`, `module-2.v2_agent-09`, `module-2.v3_agent-09`, `module-2.v3_agent-06`, `module-2.v3_agent-07`, `m3-agent-09`, `m3-agent-08`, `m2-agent-08`, `m1-agent-09`, `module-2.v3_agent-02`, `module-2.v2_agent-05`, `m1-agent-07`, `module-2.v3_agent-04`, `module-2.v2_agent-08`, `module-2.v3_agent-03`, `m2-agent-10`, `module-2.v2_agent-04`, `module-2.v3_agent-05`, `m1-agent-02`, `m1-agent-05`, `m2-agent-01`, `module-2.v2_agent-06`, `m3-agent-07`, `m3-agent-02`, `module-2.v2_agent-02`, `m2-agent-03`, `m1-agent-03`, `m2-agent-02`, `module-2.v3_agent-01`, `m2-agent-09`, `m2-agent-06`, `module-2.v2_agent-01`, `m1-agent-04`, `m2-agent-07`, `m1-agent-10`, `module-2.v2_agent-07`, `m1-agent-06`, `module-2.v3_agent-08`, `module-2.v2_agent-10`, `module-2.v2_agent-03`, `m2-agent-04`, `m1-agent-08`

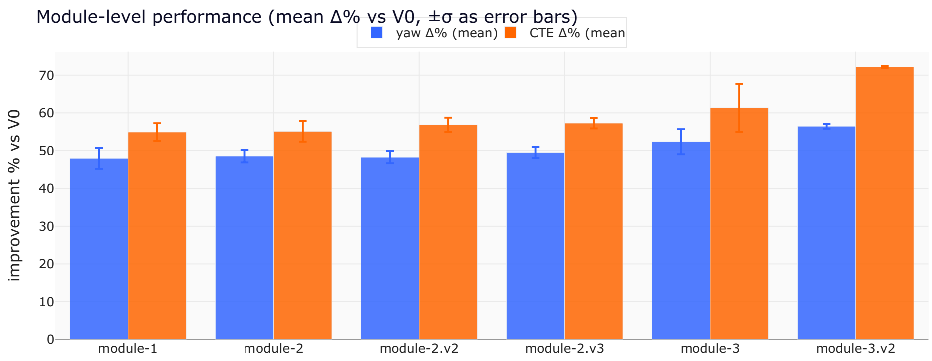
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is y = x: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

Performance grouped by which module shipped the agent. Mean Δ% over only the agents that reconstructed successfully; failures shown separately.



family	n ok / total	yaw Δ% (mean ± σ)	CTE Δ%
module-1	10/10	+48.0% ± 2.8%	+54.9
module-2	10/10	+48.6% ± 1.7%	+55.1
module-2.v2	10/10	+48.3% ± 1.6%	+56.8
module-2.v3	10/10	+49.5% ± 1.4%	+57.3
module-3	9/10	+52.3% ± 3.3%	+61.3
module-3.v2	10/10	+56.5% ± 0.6%	+72.2

2b. Token expenditure per agent

Sourced from each agent's Claude Code subagent transcript (`~/.claude/projects/<proj>/*/subagents/agent-*.jsonl`). Tokens summed across every assistant turn of the latest-mtime run per agent.

COHORT TOTAL

202.16M

MEDIAN / AGENT

3.18M

MEDIAN TURNS

64

AGENTS W/ TRANSCRIPT

60

By family

family	n	total tokens	median / agent	media
module-1	10	**23.38M**	2.38M	6
module-2	10	**25.69M**	2.24M	5
module-2.v2	10	**30.22M**	3.13M	5
module-2.v3	10	**37.55M**	3.92M	6
module-3	10	**41.38M**	3.50M	6

family	n	total tokens	median / agent	media
module-3.v2	10	**43.93M**	4.49M	

Per agent (sorted by total)

agent	family	turns	total	i
m3-agent-06	module-3	107	**8.18M**	
m3-agent-01	module-3	94	**6.78M**	
m3-agent-05	module-3	106	**6.03M**	
module-3.v2_agent-03	module-3.v2	81	**5.52M**	
module-3.v2_agent-06	module-3.v2	78	**5.16M**	
module-3.v2_agent-02	module-3.v2	76	**4.89M**	
module-3.v2_agent-01	module-3.v2	70	**4.83M**	
module-2.v3_agent-05	module-2.v3	73	**4.79M**	
module-3.v2_agent-05	module-3.v2	75	**4.75M**	
m2-agent-05	module-2	83	**4.45M**	
module-2.v3_agent-01	module-2.v3	67	**4.43M**	
module-2.v3_agent-03	module-2.v3	67	**4.26M**	
module-3.v2_agent-04	module-3.v2	64	**4.23M**	
module-3.v2_agent-09	module-3.v2	69	**4.20M**	
module-3.v2_agent-10	module-3.v2	66	**4.03M**	

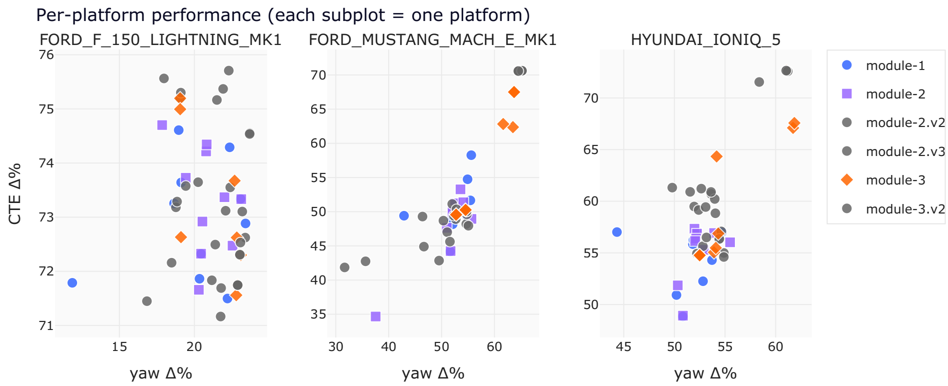
agent	family	turns	total	i
module-2.v3_agent-06	module-2.v3	73	**4.01M**	
module-2.v3_agent-02	module-2.v3	69	**3.98M**	
m3-agent-03	module-3	72	**3.96M**	
module-2.v3_agent-04	module-2.v3	67	**3.86M**	
m3-agent-08	module-3	68	**3.84M**	
m2-agent-08	module-2	76	**3.73M**	
module-3.v2_agent-07	module-3.v2	58	**3.68M**	
module-2.v2_agent-07	module-2.v2	56	**3.63M**	
module-2.v3_agent-07	module-2.v3	69	**3.61M**	
m2-agent-07	module-2	69	**3.61M**	
m1-agent-09	module-1	79	**3.54M**	
module-2.v2_agent-01	module-2.v2	64	**3.53M**	
module-2.v3_agent-10	module-2.v3	66	**3.41M**	
module-2.v2_agent-04	module-2.v2	61	**3.29M**	
module-2.v2_agent-08	module-2.v2	59	**3.19M**	
m3-agent-07	module-3	66	**3.16M**	
module-2.v2_agent-03	module-2.v2	58	**3.14M**	
module-2.v2_agent-02	module-2.v2	54	**3.11M**	
m3-agent-09	module-3	61	**3.02M**	

agent	family	turns	total
module-2.v2_agent-09	module-2.v2	57	**3.00M**
m1-agent-08	module-1	74	**2.96M**
module-2.v2_agent-05	module-2.v2	54	**2.88M**
m3-agent-10	module-3	65	**2.86M**
m3-agent-02	module-3	61	**2.83M**
module-2.v3_agent-09	module-2.v3	56	**2.72M**
m2-agent-02	module-2	68	**2.68M**
m1-agent-02	module-1	66	**2.66M**
module-3.v2_agent-08	module-3.v2	47	**2.63M**
module-2.v3_agent-08	module-2.v3	55	**2.49M**
m1-agent-10	module-1	64	**2.45M**
m1-agent-07	module-1	62	**2.41M**
module-2.v2_agent-06	module-2.v2	48	**2.37M**
m2-agent-03	module-2	52	**2.35M**
m1-agent-04	module-1	63	**2.34M**
m1-agent-05	module-1	62	**2.25M**
m2-agent-09	module-2	55	**2.14M**
m2-agent-10	module-2	53	**2.11M**
module-2.v2_agent-10	module-2.v2	42	**2.07M**

agent	family	turns	total
m1-agent-01	module-1	53	**1.88M**
m2-agent-04	module-2	45	**1.83M**
m1-agent-06	module-1	54	**1.73M**
m2-agent-06	module-2	43	**1.66M**
m1-agent-03	module-1	38	**1.17M**
m2-agent-01	module-2	41	**1.13M**
m3-agent-04	module-3	26	**740.2k**

3. Per-platform breakdown

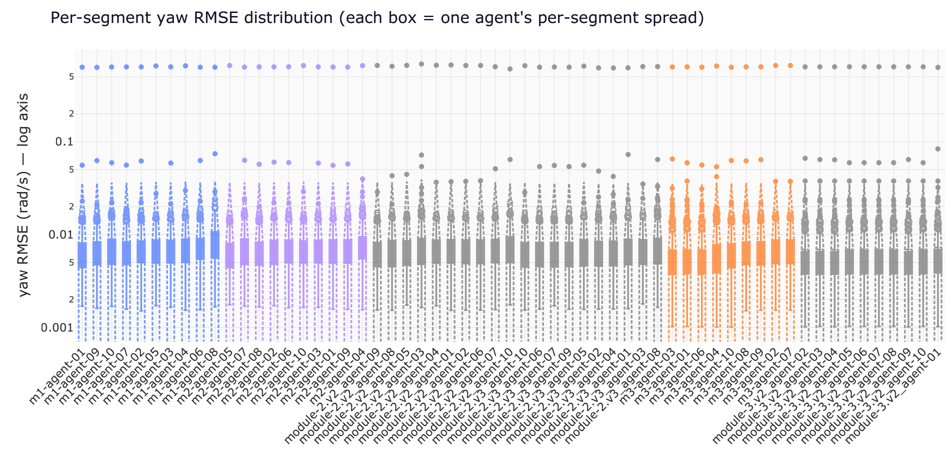
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw Δ% (mean ± σ)
FORD_F_150_LIGHTNING_MK1	59	+21.1% ± 2.2%
FORD_MUSTANG_MACH_E_MK1	59	+54.6% ± 7.2%
HYUNDAI_IONIQ_5	59	+54.5% ± 3.8%

4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status
m1-agent-01	module-1	ok
m1-agent-02	module-1	ok
m1-agent-03	module-1	ok
m1-agent-04	module-1	ok
m1-agent-05	module-1	ok

agent	family	status
m1-agent-06	module-1	ok
m1-agent-07	module-1	ok
m1-agent-08	module-1	ok
m1-agent-09	module-1	ok
m1-agent-10	module-1	ok
m2-agent-01	module-2	ok
m2-agent-02	module-2	ok
m2-agent-03	module-2	ok
m2-agent-04	module-2	ok
m2-agent-05	module-2	ok
m2-agent-06	module-2	ok
m2-agent-07	module-2	ok
m2-agent-08	module-2	ok
m2-agent-09	module-2	ok
m2-agent-10	module-2	ok
module-2.v2_agent-01	module-2.v2	ok
module-2.v2_agent-02	module-2.v2	ok

agent	family	status
module-2.v2_agent-03	module-2.v2	ok
module-2.v2_agent-04	module-2.v2	ok
module-2.v2_agent-05	module-2.v2	ok
module-2.v2_agent-06	module-2.v2	ok
module-2.v2_agent-07	module-2.v2	ok
module-2.v2_agent-08	module-2.v2	ok
module-2.v2_agent-09	module-2.v2	ok
module-2.v2_agent-10	module-2.v2	ok
module-2.v3_agent-01	module-2.v3	ok
module-2.v3_agent-02	module-2.v3	ok
module-2.v3_agent-03	module-2.v3	ok
module-2.v3_agent-04	module-2.v3	ok
module-2.v3_agent-05	module-2.v3	ok
module-2.v3_agent-06	module-2.v3	ok
module-2.v3_agent-07	module-2.v3	ok
module-2.v3_agent-08	module-2.v3	ok
module-2.v3_agent-09	module-2.v3	ok

agent	family	status
module-2.v3_agent-10	module-2.v3	ok
m3-agent-01	module-3	ok
m3-agent-02	module-3	ok
m3-agent-03	module-3	ok
m3-agent-04	module-3	ok
m3-agent-05	module-3	import_failed
m3-agent-06	module-3	ok
m3-agent-07	module-3	ok
m3-agent-08	module-3	ok
m3-agent-09	module-3	ok
m3-agent-10	module-3	ok
module-3.v2_agent-01	module-3.v2	ok
module-3.v2_agent-02	module-3.v2	ok
module-3.v2_agent-03	module-3.v2	ok
module-3.v2_agent-04	module-3.v2	ok
module-3.v2_agent-05	module-3.v2	ok
module-3.v2_agent-06	module-3.v2	ok

agent	family	status
module-3.v2_agent-07	module-3.v2	ok
module-3.v2_agent-08	module-3.v2	ok
module-3.v2_agent-09	module-3.v2	ok
module-3.v2_agent-10	module-3.v2	ok

6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
```

```
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
```

```
  "_default": {
    "a": 0.0,
    "b": 1.0,
    "c": 0.0
  }
}
```


m1-agent-06


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "best": "M2_understeer",
    "m1": {
      "a": 1.0870612234849137
    },
    "m2": {
      "L_eff": 2.4763030468069056,
      "K_us": 0.0024395167795073017
    },
    "m3": {
      "a": 0.9751496691433119,
      "b": 0.2383949727329036,
      "c": 0.00040828706448247788
    },
    "val_rmse": {
```

```
      "V0": 0.0267999381028469,
      "M1": 0.027439581036525294,
      "M2": 0.026420879339395156,
      "M3": 0.031903168689021696
    },
    "train_rmse": {
      "V0": 0.012685628695945105,
      "M1": 0.011389013310887795,
      "M2": 0.008745921115146504,
      "M3": 0.010386863485176999
    }
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "best": "M2_understeer",
    "m1": {
      "a": 0.8569324797501493
    },
    "m2": {
      "L_eff": 3.882198880475702,
      "K_us": 0.003439279945738158
    },
    "m3": {
      "a": 0.8772240685582882,
      "b": -0.031353830652592465,
      "c": -0.0036354381262862774
    },
    "val_rmse": {
      "V0": 0.01591104278075153,
      "M1": 0.011141896721881754,
      "M2": 0.0073696198273352906,
      "M3": 0.0104650790814036
    },
    "train_rmse": {
      "V0": 0.02018452928972004,
```

m1-agent-08


```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```

m1-agent-10


```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred_
    "passthrough": true
  }
}
```

m2-agent-03


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.004376953124999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
    "K_v1": 0.000785400390624989,
    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
    "a_v2": 1.1998919734409932,
    "b_v2": 3.385609342398125e-05
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
```

```
    "L": 3.0,
    "n_samples": 2042270,
    "rmse_v0": 0.017084454875116246,
    "rmse_v1": 0.009100616867655459,
    "K_v1": 0.0042265624999999985,
    "rmse_v2": 0.008741269113389604,
    "K_v2": 0.0033859035164445724,
    "a_v2": 0.9707125684779052,
    "b_v2": 0.0005066770839987745
  }
}
```


m2-agent-07


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "tau": 0.05,
    "L_eff": 3.8588781722695416,
    "K_u": 0.00314039574481963,
    "delta_bias_rad": -0.001157331673169034,
    "rmse": 0.005592721659455879
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "tau": 0.08,
    "L_eff": 2.5486304264900417,
    "K_u": 0.0015312706121448161,
    "delta_bias_rad": 4.8217638587095315e-05,
    "rmse": 0.008347407485517062
  },
  "HYUNDAI_IONIQ_5": {
    "tau": 0.05,
    "L_eff": 3.163750474852756,
    "K_u": 0.0021607848879418314,
    "delta_bias_rad": 0.0005617097326308059,
    "rmse": 0.007713638558960502
```

```
  }
}
```


m3-agent-01


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06,
    "delta0": 0.00133,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
```

```
    "tau": 0.069,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.93817,
    "L_eff": 2.8871,
    "K_us": 0.0028925,
    "tau": 0.061895,
    "delta0": 0.0,
    "use_per_segment_delta0": true
  },
  "_meta": {
    "delta0_detector": {
      "method": "yaw_rate_pred_rads_low_speed_high",
      "yr_thresh": 0.03,
      "v_thresh": 5.0,
      "min_rows": 50
    },
    "notes": "Lightning coeffs hand-set from anti-patterns.md pric
  }
}
```

m3-agent-03

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.031539710821034,
    "L_eff": 2.605202152510245,
    "K_us": 0.001860730244989818,
    "tau": 0.0627605678289049,
    "delta0_fallback": 0.00016924506921824425,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
```

```
    "g": 0.8860740528359514,
    "L_eff": 3.380345065967063,
    "K_us": 0.0032867081270640908,
    "tau": 0.05844688797895399,
    "delta0": 0.001336241196109372,
    "use_per_segment_delta0": false
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.896091628794913,
    "L_eff": 2.8008769715764292,
    "K_us": 0.0023750439062653513,
    "tau": 0.04533958799360384,
    "delta0_fallback": -0.00028615565156215293,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  }
}
```


m3-agent-06


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.0010739940929899243,
    "g": 0.8428393582368884,
    "L_eff": 3.26,
    "K_us": 0.0024711101464802783,
    "tau": 0.05493261284467375
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -9.587089484890604e-05,
    "delta0": -9.587089484890604e-05,
    "g": 0.8814740811859291,
    "L_eff": 2.22,
    "K_us": 0.001600317906539393,
```

```
    "tau": 0.08022498681872611
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 7.704255037895423e-05,
    "delta0": 7.704255037895423e-05,
    "g": 0.9863062253350734,
    "L_eff": 3.033023238719065,
    "K_us": 0.0028341308076772504,
    "tau": 0.055317906584940124
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```

m3-agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.6006343437133439,
    "delta0": 0.0012709657890603524,
    "K_us": 0.002411110430682984,
    "tau": 0.05905789911423279,
    "L_eff": 2.273362244343053,
    "loss": 0.005658956187762647,
    "n_segments_fit": 175
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.9449514060826516,
    "delta0": -2.802364842758471e-05,
```

```
    "K_us": 0.0023030236477064245,
    "tau": 0.06905315476018789,
    "L_eff": 2.3418351435366787,
    "loss": 0.008957550273019014,
    "n_segments_fit": 240
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.8635558962089673,
    "delta0": -0.0002751563706215791,
    "K_us": 0.0031040456441715804,
    "tau": 0.051877028201724164,
    "L_eff": 2.663564560241335,
    "loss": 0.009441409401572077,
    "n_segments_fit": 400
  }
}
```

m3-agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 1.2677013760203653,
    "L_eff": 4.8869672279395875,
    "K_us": 0.004638743872488176,
    "tau": 0.056604854781974856,
    "delta0_glob": 0.0012307692372779105,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.012681866822813753
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.4999996985088524,
    "L_eff": 3.7976856683656175,
    "K_us": 0.0030911056729474493,
    "tau": 0.06768316418910694,
    "delta0_glob": 0.0003094591239113909,
    "use_per_segment_delta0": true,
    "train_yaw_rmse": 0.013524463171650382
  },
```

```
  "HYUNDAI_IONIQ_5": {
    "g": 1.4070879006714407,
    "L_eff": 4.3500385986664405,
    "K_us": 0.004915645105647427,
    "tau": 0.05040893663177004,
    "delta0_glob": -0.0005219437645709744,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.009412230675873393
  }
}
```


module-2.v2_agent-02


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.9366323579288449,
    "K_us": 0.0008734759361408742,
    "b": -0.004441819259140514
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 1.1760967499879564,
    "K_us": 0.0008734571660548825,
    "b": 0.0001733616708132899
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.9342137556967759,
    "K_us": 0.0009874770996559546,
    "b": 0.002002086108887664
  },
  "TESLA_MODEL_3": {
    "k": 1.0,
    "K_us": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-04


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "d0": 0.0012435085448964291,
    "kk": 0.9364948189665847,
    "KK": 0.0032470115980856545,
    "tau": 0.0
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.97,
    "d0": -0.0005236657821362199,
    "kk": 0.933574693892375,
    "KK": 0.0029030763922389792,
    "tau": 0.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "d0": -2.675943355525215e-05,
    "kk": 1.1761809849040472,
    "KK": 0.0026070689183535123,
    "tau": 0.0
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "d0": 0.0,
    "kk": 1.0,
    "KK": 0.0,
    "tau": 0.0
  }
}
```

module-2.v2_agent-06


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9397533271985806,
    "K": 0.0008518149458872777,
    "b": -0.0044345248756529525
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 1.1808972441892185,
    "K": 0.0008794604321610935,
    "b": 0.00016328778675285131
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9350770580282316,
    "K": 0.0009883221917463027,
    "b": 0.002004778269326024
  },
  "TESLA_MODEL_3": {
    "a": 1.0,
    "K": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-08


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "G": 0.9041237868107276,
    "Kus": 0.0007955703708621087,
    "bias": -0.005088051550305564,
    "tau": 0.05804217686893913,
    "kdot": -0.012302023150477267
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "G": 1.1560312029550013,
    "Kus": 0.0010408901778361697,
    "bias": 0.0007331723308643077,
    "tau": 0.040688651274880756,
    "kdot": 0.008800778388725197
  },
  "HYUNDAI_IONIQ_5": {
    "G": 0.9700159639975359,
    "Kus": 0.001582087902935487,
    "bias": 0.00211745029940103,
    "tau": 0.04998898336470954,
    "kdot": 3.516097849258846e-05
  },
  "TESLA_MODEL_3": {
    "G": 1.0,
    "Kus": 0.0,
    "bias": 0.0,
    "tau": 0.0,
    "kdot": 0.0
  }
}
```

module-2.v2_agent-10


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.0009299897686168917,
    "g": -0.1314637793499154,
    "b": -0.004858365647164137
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 0.0007776286604362939,
    "g": 0.1760123050495998,
    "b": 0.0007151284926824161
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.0017086888155224455,
    "g": -0.02937372993329363,
    "b": 0.003098909156662854
  },
  "TESLA_MODEL_3": {
    "k": 0.0,
    "g": 0.0,
    "b": 0.0
  }
}
```


module-2.v3_agent-02


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.941941906538264,
    "K_us": 0.0037788233688761494,
    "b": -0.005450661568006636,
    "tau": -0.08207658150714123
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
    "L_eff": 2.5622977171066155,
    "K_us": 0.002745633390344498,
    "b": 0.00039676132180812505,
    "tau": -0.02283428861943423
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.0387877407606987,
    "K_us": 0.004732031059157526,
    "b": 0.0019402727789656954,
    "tau": -0.038122141286844304
  },
  "TESLA_MODEL_3": null
}
```

module-2.v3_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.96394551474604,
    "Kus": 0.003414866921617812,
    "bias": -0.005292773687032753,
    "tau": -0.06157679065853846
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.569483805164948,
```

```
    "Kus": 0.0025788290081397765,
    "bias": 0.0011230859100703557,
    "tau": -0.04709908110775255
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.008377713685593,
    "Kus": 0.005133658670161866,
    "bias": 0.0019487320168033527,
    "tau": -0.04867295794401198
  },
  "TESLA_MODEL_3": {
    "dummy": 0.0
  }
}
```


module-2.v3_agent-06


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.818759956786134,
    "Kus": 0.0039432563422251285,
    "tau": -0.05861405092290705,
    "bias": -0.004437268567086911,
    "rmse_fit": 0.0054979371675811595,
    "n": 430990,
    "converged": true
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.4997194805464074,
    "Kus": 0.0024024802542751437,
    "tau": -0.06630813663570458,
    "bias": 0.00022227158439854945,
    "rmse_fit": 0.00894166094428278,
```

```
    "n": 615175,
    "converged": true
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
    "L": 3.100074393615448,
    "Kus": 0.0035463766130646316,
    "tau": -0.05358525493540066,
    "bias": 0.001966636983251082,
    "rmse_fit": 0.008266374104865987,
    "n": 2042270,
    "converged": true
  }
}
```


module-2.v3_agent-08


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K_us": 0.004622784347956018,
    "tau": -0.05460138412592134,
    "bias": -0.004482561615678892,
    "train_mse": 3.3520085701320475e-05,
    "train_rmse": 0.005789653331704799,
    "n_samples": 430990,
    "n_segments": 164,
    "converged": true,
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  },
```

```
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K_us": 0.0008351228965168481,
    "tau": -0.09459794094766563,
    "bias": 0.00054055263346224,
    "train_mse": 0.00017605613840148047,
    "train_rmse": 0.01326861478834473,
    "n_samples": 615175,
    "n_segments": 232,
    "converged": true,
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.95,
    "K_us": 0.004646731366208564,
    "tau": -0.04946605447751251,
    "bias": 0.0018472769568295111,
    "train_mse": 7.578893405982132e-05,
    "train_rmse": 0.008705684008727937,
    "n_samples": 2042270,
    "n_segments": 785,
    "converged": true,
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
  }
}
```


module-2.v3_agent-10


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "s_d": 0.9300245172021879,
    "c_d": 0.347294150023435,
    "tau_d": -0.05773764830098028,
    "K_us": 0.003240471803333766,
    "b": -0.004299834214218103,
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
    "s_d": 1.1508056993666387,
    "c_d": 0.7498585082419895,
    "tau_d": -0.07607250144234215,
    "K_us": 0.0022785419555576775,
    "b": 0.0003167396785048121,
    "L": 2.984
  },
  "HYUNDAI_IONIQ_5": {
    "s_d": 0.9422644146648296,
    "c_d": 0.3465308670029774,
    "tau_d": -0.050503079814885346,
    "K_us": 0.002862544790766583,
    "b": 0.0022556662651586025,
    "L": 3.0
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```


module-3.v2_agent-02


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8599716540800916,
    "L_eff": 3.260802128118046,
    "K_us": 0.0033605485538025595,
    "tau": 0.058798336746219135,
    "delta0": 0.0012445223286377857,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8948751688442338,
    "L_eff": 2.218474888909999,
    "K_us": 0.002002066992546276,
    "tau": 0.06321825538415474,
    "delta0": -0.0001,
```

```
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.928157495232038,
    "L_eff": 2.890194195032293,
    "K_us": 0.002716802199963435,
    "tau": 0.051288867659338114,
    "delta0": 0.0005262026839657872,
    "use_per_segment_delta0": true
  }
}
```

module-3.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8627559173927216,
    "L_eff": 3.2661642837279934,
    "K_us": 0.003399629365556295,
    "tau": 0.05764217451751205,
    "delta0": 0.0011797692301086372,
    "delta0_fallback": 0.0011797692301086372,
    "use_per_segment_delta0": false,
    "train_rmse": 0.005285188367801501,
    "dev_rmse": 0.006580546885659461,
    "bias_spread_std": null
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.2845766366025777,
    "L_eff": 3.185078856835661,
    "K_us": 0.002783912493950955,
    "tau": 0.06245560987504152,
    "delta0": -0.0019790637549735985,
    "delta0_fallback": -0.0019790637549735985,
    "use_per_segment_delta0": true,
    "train_rmse": 0.008444302053948288,
    "dev_rmse": 0.008286227301504138,
    "bias_spread_std": null
```

```
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.9454031576502736,
    "L_eff": 2.9345636591691626,
    "K_us": 0.0028198034203513055,
    "tau": 0.0507163687871907,
    "delta0": 0.0004041780050478772,
    "delta0_fallback": 0.0004041780050478772,
    "use_per_segment_delta0": true,
    "train_rmse": 0.007930751971921808,
    "dev_rmse": 0.00660747263955304,
    "bias_spread_std": null
  }
}
```

module-3.v2_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
```

```
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
  }
}
```

module-3.v2_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
```

```
  }
}
```

module-3.v2_agent-10


```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
    "L_eff": 2.887,
    "K_us": 0.00289,
    "tau": 0.062
  }
}
```

6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

> **Clean cohort:** no agent referenced truth-derived column names inside their predict() body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
`agent_folder_exists`	60	0

format check	pass	fail
has_manifest_json	60	0
manifest_parsable	60	0
manifest_declares_predict_callable	60	0
manifest_declares_platform_support	60	0
has_predict_py	60	0
has_coeffs_json	53	7
has_report	24	36

Failure reasons across the cohort:

- import_failed — 1 agent(s)

8. Worst-of-cohort

Lowest yaw Δ%

- m1-agent-08 (+40.4%)
- m2-agent-04 (+44.6%)

Lowest CTE Δ%

- m1-agent-10 (+49.5%)
- m2-agent-07 (+49.5%)

- module-2.v2_agent-03 (+45.3%)

- m2-agent-04 (+50.0%)

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